

CHARLES MARION UELTSCHHEY

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QUALIFICATION /OBJECTIVE SUMMARY

I am an undergraduate computer science major at Mississippi State University, currently working as an undergrad researcher developing security software for private and governmental organizations. Learn more at <https://cueltschey.com>

EDUCATION

Mississippi State University, Starkville, MS

May 2026

Bachelor of Computer Science

Cumulative GPA: 3.79 GPA, President's List (2023, 2024), Dean's List (2023)

RELEVANT SKILLS

Programming Languages: C/C++(4 yrs), Bash scripting(4 yrs), Python(4 yrs), Go(1 yr), Dart(1 yr), Javascript/Typescript(2 yrs)

Proficient in: IPC, Systems Programming, Secure System Design, Software Architecture, Concurrent programming, CI CD, Cloud Computing, Firmware Reverse Engineering, Wireless Systems Design, Development Leadership

Languages: Russian (Fluent), Amharic (Intermediate)

NOTABLE SOFTWARE PROJECTS

Most my projects (36) can be found on my github: <https://github.com/cueltschey>

National Telecommunications and Information Association Software Tester User Equipment

Repo Link: <https://github.com/oran-testing/soft-t-ue>

Languages and Tools: C/C++, Python, Docker, UHD, GNU Radio

- ❖ Fully dockerized system for easy deployment on any machine
- ❖ Created custom C++ SDR channel agent for spoofing, jamming and replay attacks
 - <https://github.com/oran-testing/uu-interface-agent>
- ❖ Implemented more than 15 unique attacks on USRPs and commercial RUs
- ❖ Generated security assessments for industry professionals
 - showcased the system in multiple conferences

Wireless Compression Optimization

Repo Link: https://github.com/cueltschey/5G_dynamic_compression

Languages and Tools: C/C++, libtorch, srsRAN Project 5G RAN

- ❖ Implemented 4 new compression algorithms in the OFH of a 5G 7.2 split
- ❖ Implemented AI agent to dynamically select compression algorithms

My Own Package Manager

Repo Link: <https://github.com/cueltschey/some-pkgmgr>

Languages and Tools: Go

- ❖ Multiple threading for parallel downloads
- ❖ Significantly faster than apt, but a slightly less stable

Convolutional Neural Network for Identifying Microorganisms

Repo Link: <https://github.com/cueltschey/MicrobeNet>

Languages and Tools: Python, Selenium, TensorFlow, Google cloud (training platform)

- ❖ Applied scharr image convolution before training and saw 5% efficiency increase
- ❖ Accuracy result of 90% surpassed the current standard of 70% in existing literature

WORK EXPERIENCE

MSU Wireless Research Lab

January 2024 - Current

NTIA project lead developer

- ❖ Managed a team of 7 undergraduates in developing for several wireless projects
- ❖ Worked 40 hours per week designing and creating software alongside professors and research assistants
- ❖ Collaborated to write 4 published research papers
- ❖ Attended several conferences to present software projects to the community

Self Employed

November 2023 - Current

Freelance Developer